

Lederberg and his colleague E. L. Tatum set out to determine whether a sexual process might occur in bacteria. The bacterial species he used in the experiments was *Escherichia coli*. This was fortuitous, as it turned out, because *E. coli* often contains a special kind of conjugative plasmid that has the ability to insert itself into the cell's chromosome. Once this happens, the donor cell can transfer to a recipient not only plasmid genes but also large numbers of chromosomal genes.

Lederberg worked with two different nutritional mutants of *E. coli*. One strain required biotin and methionine to grow; the other strain required threonine and leucine. Lederberg mixed the two strains together and then attempted to grow them without supplying any of the four nutrients. His hypothesis was that any cell able to grow without the four nutrients would have all four genes intact, and would thus have received the functioning genes from the other strain and incorporated them into its chromosome. The incorporation of the genes in this manner is called genetic recombination.

As he predicted, Lederberg's experiment yielded cells that did not require any of the nutrients to grow. In a second set of experiments, Lederberg showed that cell-to-cell contact was necessary for genetic recombination to occur. Over several years, he and other scientists discovered the mechanics of the entire process that we now call conjugation.

Antibiotic Resistance

From the human perspective, one of the significant consequences of a bacterium's ability to pass genetic information along to other cells via conjugation is its link to the widespread incidence of antibiotic resistance. The genes that encode for resistance to a variety of antibiotics like penicillin and tetracycline are commonly found on plasmids. When a population of susceptible bacteria is exposed to a given antibiotic, most of them will be killed. However, if the population contains cells with conjugative plasmids bearing the genes for resistance, they can rapidly spread the trait throughout the population. These plasmids are large and are often promiscuous, so that transfer of antibiotic resistance genes need not be restricted to cells of like species. In some cases, this has resulted in disease-causing bacteria that are resistant to almost every antibiotic available. For instance, antibiotic resistant tuberculosis bacteria are a significant public health threat in some metropolitan areas. SEE ALSO ANTIBIOTIC RESISTANCE; *ESCHERICHIA COLI* (*E. COLI* BACTERIUM); PLASMID; RECOMBINANT DNA; TRANSDUCTION; TRANSFORMATION.

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biodiversity degree of
variety of life

Conservation Biology: Genetic Approaches

Conservation biology is a multidisciplinary field dedicated to protecting global **biodiversity** and critical habitats. It incorporates biological approaches such as ecology, evolution, and behavior studies, as well as other disciplines, such as political science, law, economics, and cultural anthropology. One of the major goals of conservation biology is preserving critical habitats and the species that inhabit them. Genes can tell us something about how a particular habitat is used by species and populations. Genetic approaches are also used to identify and classify organisms and evaluate the extent of genetic diversity within a particular population.

Categories of Threatened Populations

The International World Conservation Union (IUCN) provides definitions of terms used to describe the status of a species in the wild, based on a number of factors, including the size of a particular population, whether the population is declining in number, and, if so, the extent to which the trend will continue, as well as the threats the population faces. Genetic approaches can play an essential role in helping to evaluate populations, species, and species designations.

The following categories currently cover the range of definitions for species status according to IUCN: Extinct, Extinct in the Wild, Critically Endangered, Endangered, Vulnerable, Lower Risk, and Data Deficient.

Threatened. Populations are considered critically endangered, endangered, or vulnerable when there is considerable concern, based on available evidence or a high level of uncertainty, that the population will survive. With any of these classifications, the species or population of concern is considered to be facing a high to very high risk of extinction in the wild.

Conservation Genetics Applications

The practical applications of conservation genetics include analyzing fragmented populations in nature, determining units of conservation in nature, and monitoring captive populations. In general, conservation genetics integrates these types of information on particular species and populations to help prioritize areas for conservation.

Conservation genetics also plays a major role in guiding relocation and reintroduction efforts, in prioritizing species for conservation, and in designing captive-breeding programs. Identifying natural units based on systematics and population genetics allows researchers and wildlife officials to track organisms in the wild and in zoos, and it lets them identify parts or products of endangered and threatened organisms that are used in illegal trade. Conservation geneticists may use genetic techniques to determine, for example, if certain individuals in the pet-trade were illegally taken from the wild versus bred from permitted captive breeding programs.

Some of the most common issues addressed by genetic techniques in conservation are those confronting small or fragmented populations. Genetic approaches in these cases allow researchers to assess the variability in these populations, as well as to assess whether there is any history or

future danger of loss of genetic variability. Genetics can help conservation biologists do viability analyses (tests of how likely that a population will survive over time) by testing **hypotheses** concerning how long genetic variation might persist into the future. This might be done by examining current levels of genetic variation in a species or population, and integrating these pieces of information with demographic and life history models to examine what happens to genetic variation over time.

The use of a conservation genetics approach may be an effective way for assessing the status of populations and species in the wild. Populations that decrease in number while becoming increasingly fragmented by loss of habitat in the wild can experience a loss of genetic variation that could have a severe impact on their fitness and survival. Conservation genetics permits scientists to assess the impacts of habitat fragmentation and loss in the wild using both theoretical and empirical methods. Results from these studies allow managers to evaluate the viability of populations and design protected areas for conservation.

Sometimes conservation initiatives are also concerned with the translocation or reintroduction of animals to areas where they have been extirpated or severely depleted. Such reintroduction or translocation measures require a detailed understanding of the genetics of the populations being reintroduced in order to ensure there is compatibility between populations as well as to maximize genetic variation and minimize the chance of inbreeding among related animals.

Determining the extent of genetic variation among captive populations in zoological parks and botanical gardens is also essential, because captive populations must have sufficient genetic variation so that they persist into the future without suffering from reduced fitness due to inbreeding and other effects associated with small populations. In some cases, captive populations may be viewed as a source for improving genetically or numerically depleted wild populations. However, they must be managed to minimize the effects of inbreeding. Accredited zoos, aquariums, and botanical gardens work to manage populations and establish conservation programs that strive to carefully manage the breeding of a species in captivity. The primary goal is to maintain a healthy and self-sustaining captive population that is both genetically diverse and demographically stable. Captive breeding specialists usually attempt to maximize the genetic health of a population by reconstructing pedigrees of the animals in the captive populations in order to understand and minimize how much inbreeding might occur.

The Tools of Conservation Genetics

The technique that revolutionized modern molecular genetics is the polymerase chain reaction (PCR). PCR has had major implications for conservation genetics. This technique allows the amplification of minute amounts of DNA, which can then be used for analysis. Amplification is critical for the study of endangered species, because biological samples may be obtained from nontraditional sources, such as hair, feathers, sloughed skin, or feces from which only small amounts of DNA are generally available. Once DNA has been obtained, conservation geneticists are able to use a wide arsenal of tools to characterize the genetics of endangered and threatened species and populations.

hypotheses testable statements



When conservation genetics is used to decipher the evolutionary relationships among species, DNA sequence comparisons are often made. Sequencing a region of a gene and properly analyzing the data may lead to novel findings. Based on analyzing underlying genetic variation, there could be evidence to suggest that a revision of numbers of species might be warranted. What was once considered a single species with two populations, for example, might actually merit consideration for separate species status. The level of genetic differentiation detected could have significant implications for how they are protected in the wild and the measures that must be taken by local, state, and federal authorities. Alternatively, the data may indicate that these populations are not sufficiently genetically different to merit separate species designations. DNA sequences are also used to aid in diagnosing natural units for conservation in the wild. Detecting fixed nucleotide characters among DNA sequences between well-sampled populations can provide sufficient evidence for defining units of conservation that potentially merit separate species status under the Phylogenetic Species Concept. Several studies have used DNA sequence polymorphism shared by certain units to the exclusion of other groups to unequivocally define or diagnose species.

Population-level analyses use DNA sequences as well as another set of molecular markers, called microsatellites, a type of repetitive DNA element. Microsatellites are used to address many conservation genetics questions. They are short, tandem-repeated motifs of DNA sequences, such as a dinucleotide repeat (e.g., $(AT)_n$), that usually vary in the number of repeats in a particular stretch of DNA. They are distributed throughout the genomes of plants and animals, are inherited in a Mendelian fashion, and have been found to be highly **polymorphic**. These genetic markers have proven to be useful in population studies for such purposes as estimating gene flow between populations, describing the genetic variation within and between populations, and examining the effects of hybridization between species. They are used in pedigree analysis to identify individuals based on a DNA sample, and they are used to decipher mating strategies and degrees of relatedness among members of a population.

polymorphic occurring
in several forms

Implication of Genetics for Conservation in the Wild

In the wild, populations that once were large and widespread are increasingly being reduced to small and fragmented isolates due to human activities. Habitat loss and fragmentation trigger processes that further threaten populations. Small populations often face greater demographic and genetic risks relative to large populations. When populations become fragmented and small, the genetic diversity of a population may be greatly affected. Conservation geneticists focus on the impact of such severe reductions, called bottlenecks, on endangered species.

When a bottleneck occurs, there is an increased chance of breeding among close relatives. This is termed inbreeding, and it may result in a reduction in fitness due to the expression of deleterious genes, in a process known as inbreeding depression. Inbreeding and the loss of genetic variation in small populations can lead to a genetically reduced or homogeneous population that is more sensitive to diseases and to the effects of habitat alteration. The interaction between genetic and demographic declines has been termed “extinction vortex.” We include below several real examples of the use of genetics in conservation biology.

Bottlenecks, Cheetahs, and Right Whales. In small populations, inbreeding depression may be more common because random mating is less likely and breeding among related animals may have a greater cumulative effect. Low genetic diversity and inbreeding is not always deleterious, and some small populations may be stable while permanently maintaining low levels of genetic diversity. In general, however, avoidance of inbreeding is a major goal in the management of small populations, since it has been shown to cause a reduction in fitness in captive populations of endangered species.

Perhaps the most famous case of a putative bottleneck being examined in conservation genetics is the cheetah, as examined by Stephen J. O'Brien, a molecular geneticist at the National Cancer Institute. In this 1980s study, cheetahs were shown to have extremely low levels of genetic diversity, which the researchers attributed to a bottleneck that happened less than ten thousand years ago and that may have left only a few females alive. The bottleneck was so extreme that even the usually highly diverse genes of our immune system, genes of the major histocompatibility complex, showed amazingly low levels of diversity. The extreme loss of genetic diversity was attributed to difficulties associated with the species' breeding in captivity and in the wild, abnormal sperm counts, and susceptibility to disease. While it remains controversial whether cheetah populations went through a bottleneck and the extent to which reproductive issues can be attributed to reduced genetic variation, the example remains one of the most prominent in the field of conservation.

Other species, such as the North Atlantic right whale, have faced demographic decline and extremely low levels of genetic diversity since the end of legal commercial hunting for whales at the beginning of the twentieth century. The North Atlantic right whale has maintained a low level of genetic diversity since the 1930s, but recent studies suggest that some additional genetic variation may eventually be lost.

Units of Conservation and DNA Sequences. Using genetic data to evaluate or define species and/or units of conservation can also lead to novel findings and enhance conservation management. Right whales are found in the North Atlantic, North Pacific, and southern oceans. For over a hundred years, they have been considered as two species—one in the north (in the Pacific and Atlantic Oceans) and one in the south. DNA diagnosis methods have corroborated that the southern right whales are distinct from all the others. However, the DNA data also clearly demonstrate that the North Pacific and North Atlantic whales are distinct from each other and warrant distinct species status. The ramifications for the conservation plan of these whales have been taken into consideration by the appropriate management authorities, who have developed a revised plan for naming and protecting three distinct species of right whales.

DNA Detectives and Endangered Species. As discussed above, conservation genetics can aid in the identification of endangered and threatened animals that are traded illegally as commercial products. Researchers at the Wildlife Conservation Society used species-identification methods to detect caiman crocodile tissue in leather products and thus thwart their illegal importation into the United States. Other scientists have used species identification methods to detect whale meat in Japanese fish markets and thus have had an impact on the policing of whale hunting.



Rob DeSalle, a curator at The American Museum of Natural History, and colleagues have recently used species identification methods to verify or reject the labeling of caviar origin. Such tests have been instrumental in getting sturgeons (the source of caviar) listed on the Convention on International Trade in Endangered Species of Wild Fauna and Flora. Prior to the test's development, there was no way to verify the contents of a container of caviar, leading authorities to be wary of prosecuting the illegal importation of caviar. With the development of species-identification procedures based on analyzing the DNA from single caviar eggs, enforcement of importation regulations became possible. SEE ALSO CONSERVATION BIOLOGIST; DNA PROFILING; GENE FLOW; POPULATION BOTTLENECK; REPETITIVE DNA ELEMENTS.

Howard C. Rosenbaum and Rob DeSalle

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Conservation Geneticist

A career path in conservation biology can be taken in many different directions, many of which are not centered around genetics. In general, the science of conservation biology draws upon many traditional academic disciplines, combining them in a science dedicated to maintaining Earth's **biodiversity**. In recent years, however, genetics has taken on an increasingly important role in the field, for a genetic approach can be used to address questions concerning conservation of populations, species, and their habitats.

A Variety of Career Directions

Conservation biologists, and more specifically conservation geneticists or molecular ecologists, are often multi-disciplinary scientists who combine training in ecology, evolutionary biology, conservation, and molecular biology. This unique combination of skills enables these scientists to design field studies aimed at collecting biological specimens needed for research involving genetics, or to advise others who are planning such studies.

Genetics-based conservation involves both research and the implementation of that research's findings. Some conservation geneticists focus their research on identifying natural population units for conservation based on genetic criteria, in an effort to maximize genetic diversity. Others seek to establish **taxonomical** or population priorities for conservation efforts. Alternatively, a conservation geneticist may monitor trade in endangered species, guide captive-breeding programs, or work on the re-introduction of selected species to habitats from which they are in danger of disappearing. Results from their studies are often central for management decisions regarding the viability and protection of threatened or endangered populations and the designation of critical habitats for conservation.

The discovery of the polymerase chain reaction has enabled conservation biologists to study **endangered** or **threatened** species and

biodiversity degree of variety of life

taxonomical derived from the science that identifies and classifies plants and animals

endangered in danger of extinction throughout all or a significant portion of a species' range

threatened likely to become an endangered species



their habitats in ways that were previously unimaginable. The use of non-invasively collected samples, such as feces or hair, enables the conservation geneticist to acquire an extraordinary amount of information about life history, **demography**, distribution, and diversity of rare species without directly observing them. In addition to such studies, with research drawn from skilled geneticists, ecologists, and behavioral biologists, conservation geneticists are now faced with a new set of high-tech approaches derived from the emerging field of genomics. The challenge for these biologists will be to evaluate whether these advanced technologies can be used effectively to promote biodiversity, conservation, and habitat protection.

Conservation biologists are likely to lead projects such as this whooping crane breeding program located in Patuxent, Maryland.

demography aspects of population structure, including size, age distribution, growth, and other factors

Becoming a Conservation Biologist

To begin a career in conservation biology, a student should expect to carry out advanced study in one or more of the relevant sciences at the graduate level. Many researchers have completed doctorates in biology, genetics, or conservation, followed by several years of post doctoral training. Projects that are initiated as part of graduate work may develop into entire research programs in this field.

The key to success for a conservation geneticist goes far beyond designing and carrying out a research program. Equally important is how the program implements the results of that research, and it is often this

implementation that is used as a benchmark in evaluating the program's value. Research in this discipline may be carried out at academic centers, museums, nongovernmental organizations, and government institutions.

The rewards of a career in conservation biology are many. First, the multidisciplinary approach affords a researcher the chance to develop a unique background and breadth of diverse skills. Second, the work has a direct and clearly visible value for biodiversity conservation programs. Salaries for a conservation geneticist depend on the researcher's level of education and the type of institution in which he or she works, but most are comparable to the salaries offered to biology professors teaching at the university level. SEE ALSO CONSERVATION BIOLOGY: GENETIC APPROACHES; POLYMERASE CHAIN REACTION.

Howard C. Rosenbaum and Rob DeSalle

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Crick, Francis

British Biophysicist 1916–

Francis Crick is the co-discoverer, with James Watson, of the structure of DNA. He has remained a significant contributor to theoretical biology since that discovery.

Education and Training

Crick was born in Northampton, England, in 1916. He studied physics at University College in London until the outbreak of the Second World War. He then joined the British Admiralty Research Laboratory, where he contributed to the development of radar for tracking enemy planes, and magnetic mines used in naval warfare.

During this time, Crick read *What is Life?*, a book by the physicist Erwin Schrödinger. Schrödinger's book popularized the work of physicist Max Delbrück, who had begun to apply the analytical tools of physics to inquire what a gene was and how it might behave. Like many other physicists at that time, Crick was excited by Delbrück's approach, and turned his attention to biochemistry and biological physics. While he knew a great deal of physics, he knew very little chemistry or biology at that time. In 1949 he began research at the Cavendish Laboratory in Cambridge, England, using **X-ray crystallography** to study the three-dimensional structures of proteins. At that time, Crick wrote that he was interested in "the borderline between the living and the nonliving, as typified by, say, proteins, viruses, bacteria and the structure of chromosomes. The eventual goal, which is somewhat remote, is the description of these activities in terms of their structure, i.e., the spatial distribution of their constituent atoms" (Judson, 88).

X-ray crystallography
use of X rays to determine the structure of a molecule

The Structure of DNA

Almost ten years earlier, it had been shown that genes encode proteins, but the chemical nature of the gene remained unknown. Genes were presumed to be composed of DNA (deoxyribonucleic acid), at least in part, but how DNA might encode hereditary information, and whether it acted alone or in partnership with proteins, was a complete mystery. Crick saw that the solution to the mystery lay in discovering the structure of DNA, whose linearity he guessed corresponded to the linear **amino acid** chains of which proteins are made.

In 1951 a 23-year-old American named James Watson joined the Cavendish Laboratory. Watson and Crick got along well, and they decided to work together on the structure of DNA. DNA was known to be composed of nucleotide subunits, each of which had a sugar (deoxyribose), a phosphate, and a nitrogenous base. The sugars were known to alternate with phosphates to make long strands, off of which the bases projected. The bases came in four types: adenine, thymine, cytosine, and guanine (A, T, C, and G). Shortly before Crick and Watson began to collaborate, American biochemist Erwin Chargaff had discovered that across a wide range of species, the amount of adenine in an organism's DNA always equaled the amount of thymine, and the amount of cytosine always equaled the amount of guanine.

Crick and Watson proceeded to build models of the nucleotides, which they attempted to fit together in accordance with what was known from experimental data. The most important data came from X-ray images of DNA that had been generated by Rosalind Franklin, who also worked at the Cavendish. Using this information, they constructed a model in which the two sugar-phosphate strands wind around each other to form a double helix, their bases projecting inward, like the stair treads of a broad spiral staircase. The two strands are held together and stabilized by the hydrogen bonding between the bases across the interior. These weak chemical attractions, they discovered, are strongest when adenine projects across to meet a thymine, and guanine a cytosine, explaining the ratios discovered by Chargaff. They published their model in 1953. Watson and Crick received the Nobel Prize in physiology or medicine in 1962 for this work, along with Maurice Wilkins of the Cavendish Lab.

After the publication of DNA's structure, Crick turned his attention to understanding the coding function of DNA. He and Watson proposed that the order of bases in a gene encoded the order of amino acids in a protein. Over the next decade, the details of this insight were worked out by a large group of scientists, including Crick, Watson, Sydney Brenner, George Gamow, Seymour Benzer, Marshall Nirenberg, and Har Gobind Khorana. As part of this work, Crick hypothesized the existence of an "adaptor" that intervened between DNA and amino acids. This led to the discovery of messenger RNA and transfer RNA, which serve this function.

Later Work

Crick received his Ph.D. in 1954. He remained with the Medical Research Council at the Cavendish Laboratory, and became head of the Division of Molecular Genetics in 1962, continuing to work closely with Sydney Brenner. He turned his attention to embryology in the mid-1960s, and in 1975 he moved to the Salk Institute in La Jolla, California, to pursue

amino acid a building block of protein



neurobiology, an interest that had vied with molecular biology from the very beginning of his career. At the Salk Institute, in collaboration with Christof Koch, he studies the neural correlates of conscious visual experience, seeking to understand how neuron firing patterns correspond to the conscious experience of seeing. SEE ALSO DELBRÜCK, MAX; DNA; DNA STRUCTURE AND FUNCTION, HISTORY; RNA; WATSON, JAMES.

Richard Robinson

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gametes reproductive cells, such as sperm or eggs

Crossing Over

Crossing over, or recombination, is the exchange of chromosome segments between nonsister chromatids in meiosis. Crossing over creates new combinations of genes in the **gametes** that are not found in either parent, contributing to genetic diversity.

Homologues and Chromatids

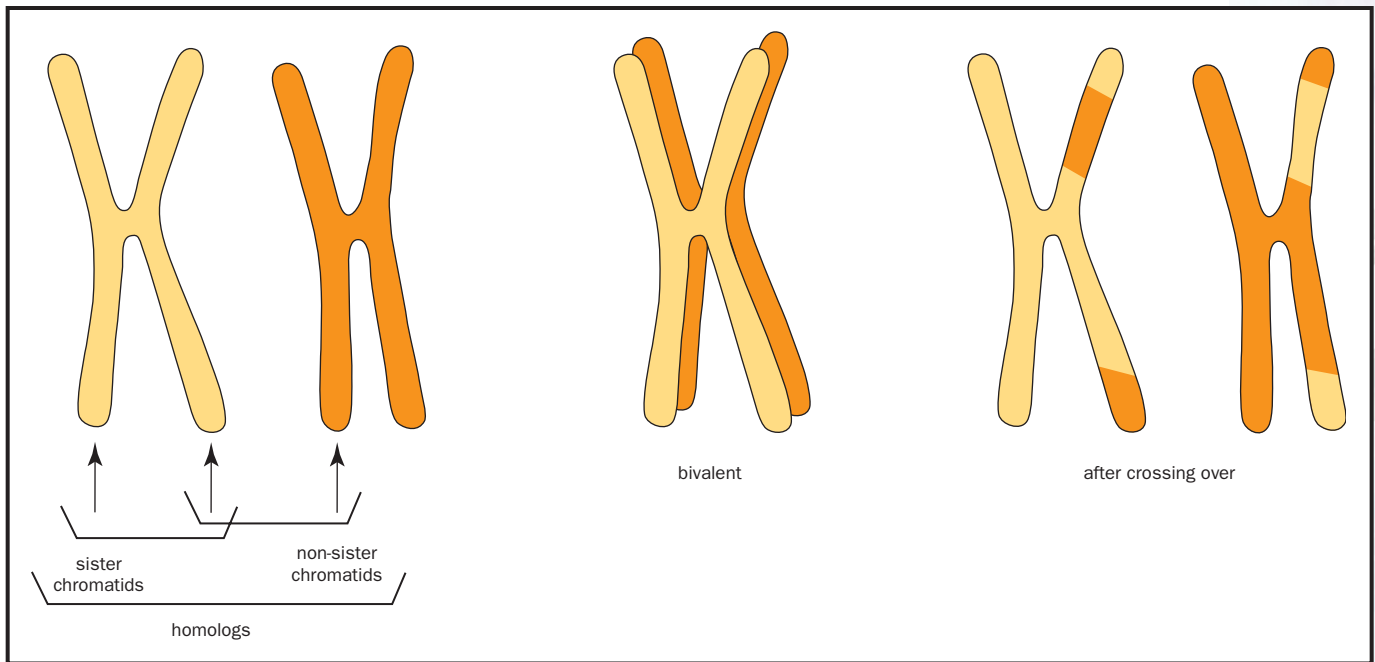
All body cells are diploid, meaning they contain pairs of each chromosome. One member of each pair comes from the individual's mother, and one from the father. The two members of each pair are called homologues. Members of a homologous pair carry the same set of genes, which occur in identical positions along the chromosome. The specific forms of each gene, called alleles, may be different: One chromosome may carry an allele for blue eyes, and the other an allele for brown eyes, for example.

Meiosis is the process by which homologous chromosomes are separated to form gametes. Gametes contain only one member of each pair of chromosomes. Prior to meiosis, each chromosome is replicated. The replicas, called sister chromatids, remain joined together at the centromere. Thus, as a cell starts meiosis, each chromosome is composed of two chromatids and is paired with its homologue. The chromatids of two homologous chromosomes are called nonsister chromatids.

Meiosis occurs in two stages, called meiosis I and II. Meiosis I separates homologues from each other. Meiosis II separates sister chromatids from each other. Crossing over occurs in meiosis I. During crossing over, segments are exchanged between nonsister chromatids.

Mechanics of Crossing Over

The pairing of homologues at the beginning of meiosis I ensures that each gamete receives one member of each pair. Homologues contact each other along much of their length and are held together by a special protein structure called the synaptonemal complex. This association of the homologues may persist from hours to days. The association of the two chromosomes is called a bivalent, and because there are four chromatids involved it is also called a tetrad. The points of attachment are called chiasmata (singular, chiasma).



The pairing of homologues brings together the near-identical sequences found on each chromosome, and this sets the stage for crossing over. The exact mechanism by which crossing over occurs is not known. Crossing over is controlled by a very large protein complex called a recombination node. Some of the proteins involved also play roles in DNA replication and repair, which is not surprising, considering that all three processes require breaking and reforming the DNA double helix.

One plausible model supported by available evidence suggests that crossing over begins when one **chromatid** is cut through, making a break in the double-stranded DNA (recall that each DNA strand is a double helix of nucleotides). A nuclease enzyme then removes nucleotides from each side of the DNA strand, but in opposite directions, leaving each side with a single-stranded tail, perhaps 600 to 800 nucleotides long.

One tail is then thought to insert itself along the length of one of the nonsister chromatids, aligning with its complementary sequence (i.e., if the tail sequence is ATCCGG, it aligns with TAGGCC on the nonsister strand). If a match is made, the tail pairs with this strand of the nonsister chromatid. This displaces the original paired strand on the nonsister chromatid, which is then freed to pair with the other single-stranded tail. The gaps are filled by a DNA **polymerase enzyme**. Finally, the two chromatids must be separated from each other, which requires cutting all the strands and rejoining the cut ends.

During crossing over, homologous chromosomes pair up to form a bivalent. Nonsister chromatids exchange segments.

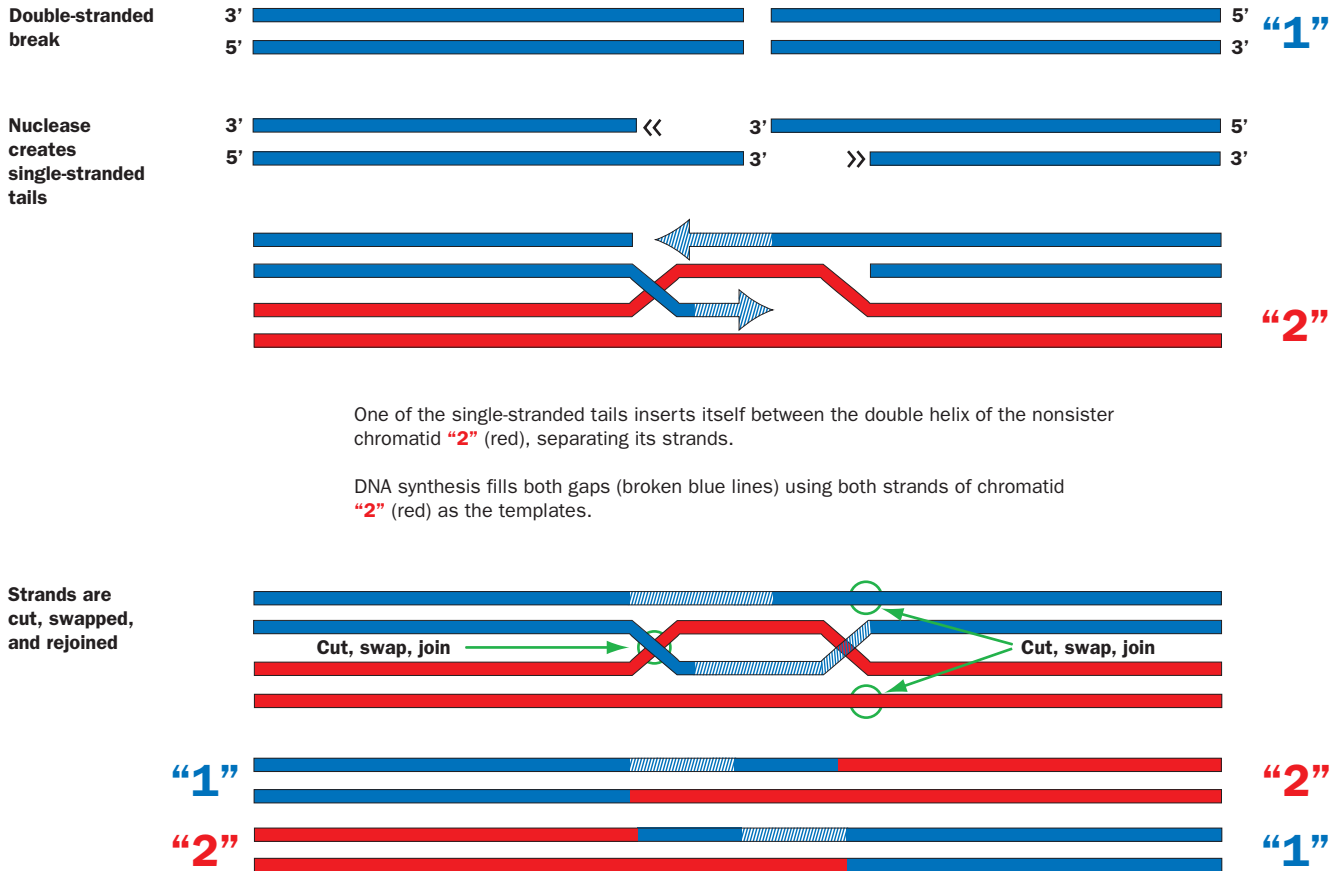
chromatid a replicated chromosome before separation from its copy

polymerase enzyme enzyme complex that synthesizes DNA or RNA from individual nucleotides

The Consequences of Crossing Over

A chiasma occurs at least once per chromosome pair. Thus, following crossing over, at least two of the four chromatids become unique, unlike those of the parent. (Crossing over can also occur between sister chromatids; however, such events do not lead to genetic variation because the DNA sequences are identical between the chromatids.) Crossing over helps

PROPOSED MECHANISM FOR CROSSING OVER



The exact mechanism for crossing over is unknown. It is believed to involve cutting of one chromatid, removal of nucleotides from both strands, insertion of one strand into the matching region of the nonsister chromatid, and further cutting, swapping of segments, and rejoining of the strands. Adapted from <<http://www.ultranet.com/~jkimball/Biology/Pages/c/crossingover.html>>.

to preserve genetic variability within a species by allowing for virtually limitless combinations of genes in the transmission from parent to offspring.

The frequency of recombination is not uniform throughout the genome. Some areas of some chromosomes have increased rates of recombination (hot spots), while others have reduced rates of recombination (cold spots). The frequency of recombination in humans is generally decreased near the centromeric region of chromosomes, and tends to be greater near the telomeric regions. Recombination frequencies may vary between sexes. Crossing over is estimated to occur approximately fifty-five times in meiosis in males, and about seventy-five times in meiosis in females.

X-Y Crossovers and Unequal Crossovers

The forty-six chromosomes of the human diploid genome are composed of twenty-two pairs of autosomes, plus the X and Y chromosomes that determine sex. The X and Y chromosomes are very different from each other in their genetic composition but nonetheless pair up and even cross over during meiosis. These two chromosomes do have similar sequences over a small portion of their length, termed the pseudoautosomal region, at the far end of the short arm on each one.

The pseudoautosomal region behaves similarly to the autosomes during meiosis, allowing for segregation of the sex chromosomes. Just proximal to the pseudoautosomal region on the Y chromosome is the *SRY* gene (sex-determining region of the Y chromosome), which is critical for the normal development of male reproductive organs. When crossing over extends past the boundary of the pseudoautosomal region and includes this gene, sexual development will most likely be adversely affected. The rare occurrences of chromosomally XX males and XY females are due to such aberrant crossing over, in which the Y chromosome has lost—and the X chromosome has gained—this sex-determining gene.

Most crossing over is equal. However, unequal crossing over can and does occur. This form of recombination involves crossing over between non-allelic sequences on nonsister chromatids in a pair of homologues. In many cases, the DNA sequences located near the crossover event show substantial sequence similarity. When unequal crossing over occurs, the event leads to a deletion on one of the participating chromatids and an insertion on the other, which can lead to genetic disease, or even failure of development if a crucial gene is missing.

Crossing Over as a Genetic Tool

Recombination events have important uses in experimental and medical genetics. They can be used to order and determine distances between **loci** (chromosome positions) by genetic mapping techniques. Loci that are on the same chromosome are all physically linked to one another, but they can be separated by crossing over. Examining the frequency with which two loci are separated allows a calculation of their distance: The closer they are, the more likely they are to remain together. Multiple comparisons of crossing over among multiple loci allows these loci to be mapped, or placed in relative position to one another.

Recombination frequency in one region of the genome will be influenced by other, nearby recombination events, and these differences can complicate genetic mapping. The term “interference” describes this phenomenon. In positive interference, the presence of one crossover in a region decreases the probability that another crossover will occur nearby. Negative interference, the opposite of positive interference, implies that the formation of a second crossover in a region is made more likely by the presence of a first crossover.

Most documented interference has been positive, but some reports of negative interference exist in experimental organisms. The investigation of interference is important because accurate modeling of interference will provide better estimates of true genetic map length and intermarker distances, and more accurate mapping of trait loci. Interference is very difficult to measure in humans, because exceedingly large sample sizes, usually on the order of three hundred to one thousand fully informative meiotic events, are required to detect it. SEE ALSO DNA POLYMERASES; DNA REPAIR; LINKAGE AND RECOMBINATION; MEIOSIS; MENDEL, GREGOR.

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loci sites on a chromosome (singular, locus)



nucleotides the building blocks of RNA or DNA

complementary matching opposite, like hand and glove

electrophoresis technique for separation of molecules based on size and charge

Cycle Sequencing

DNA sequencing is the process of determining the order of **nucleotides** on a segment of DNA. Cycle sequencing is a method used to increase the sensitivity of the DNA sequencing process and permits the use of very small amounts of DNA starting material. This is accomplished by using a temperature cycling process similar to that employed in the polymerase chain reaction.

The Chain Termination Method

The most popular approach to sequencing is the chain termination method, developed in 1977 by Fred Sanger. This technique makes use of a DNA synthesis reaction and a unique form of a base, called a dideoxynucleotide, that lacks the 3' hydroxyl chemical group involved in forming the link between nucleotides in a DNA chain. A dideoxynucleotide can be added to a growing chain, but, once incorporated, no further nucleotides can be linked to it. Thus, chain growth is terminated.

In a reaction, the concentration of dideoxynucleotides is optimized so that all possible chain lengths are generated. In automated DNA sequencing, the newly formed chain fragments are marked with four fluorescent dyes, each of which corresponds to one of the four DNA nucleotides (A, C, T, and G). The fragments are then separated by a technique known as gel electrophoresis, during which the dyes are excited and detected by the automated DNA sequencing instrument. In this way, the identity of each successive terminating nucleotide is determined, revealing the sequence of the entire chain.

In a sequencing reaction, all the components needed for the synthesis of DNA are present. These include the DNA to be sequenced (the template) and a short (17 to 28 bases in length), single-stranded piece of DNA (the primer), which attaches itself to a specific site on the template and acts as a starting point for the synthesis of a new DNA strand.

In both a DNA synthesis reaction and in the chain termination method of DNA sequencing, one strand of the template DNA is copied to form a new, **complementary** strand. An A base on the template strand, for example, directs the addition of its complementary base T into the growing chain. Likewise, a C base on the template strand directs the addition of its complementary base G into the corresponding position of the new chain. During chain growth, an enzyme called DNA polymerase links one nucleotide to the next, extending the new strand until it either reaches the end of the template strand, or, in DNA sequencing, until a dideoxynucleotide is incorporated.

The Cycle Sequencing Technique

In cycle sequencing, a reaction is taken through several steps designed to prepare the template for copying, allow for initiation of DNA synthesis, and generate the terminated DNA chains needed for **electrophoresis** and sequence determination. The first step in the process is called denaturation, in which double-stranded template DNA is converted into its single-stranded form. This is accomplished by heating the template to between 94 °C and 98 °C, a temperature high enough to break the hydrogen bonds between the complementary bases holding the two strands together.

As a single-stranded molecule, the template's bases are now exposed, and are free to interact with the sequencing primer. The primer, in a step called annealing, locates and attaches itself to its complementary site on the template. Thus, Ts bind to As and Cs bind to Gs. However, primer annealing will only occur at a temperature where **hydrogen bonds** can form between the primer and **template** strands, usually between 40 °C and 65 °C. Because the high temperature used for the **denaturation** step in each cycle would destroy most DNA polymerase enzymes, a special heat-stable enzyme must be used in the annealing stage, one that remains active even after repeated exposure to very high temperatures. The enzyme most commonly used at this point in cycle sequencing is *Taq*, isolated from *Thermus aquaticus*, a bacterium that lives in the hot springs of Yellowstone National Park.

In the final step of the reaction, DNA polymerase extends the annealed primer by sequentially adding on to its end bases that are complementary to those on the template. It is during this extension step of a DNA sequencing reaction that random incorporation of a dideoxynucleotide can occur, terminating chain growth. All three of these steps, taken together, represent one round, or cycle, of a DNA synthesis reaction. By repeating a cycle over and over again, the amount of each fragment made in the reaction can be substantially increased. Since each fragment carries fluorescent dyes, increasing the number of copied fragments also increases the strength of the fluorescent signal. Cycle sequencing, therefore, greatly improves the sensitivity of the sequencing reaction, and even very small amounts of starting DNA sample can be used as template. **SEE ALSO** AUTOMATED SEQUENCER; GEL ELECTROPHORESIS; POLYMERASE CHAIN REACTION; SANGER, FRED; SEQUENCING DNA.

Frank H. Stephenson and Maria Cristina Abilock

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Cystic Fibrosis

Cystic fibrosis (CF) is one of the most common genetic diseases and one of the best known to the general public. There are approximately one thousand new cases of CF in the United States every year, and approximately thirty thousand people in the country are currently affected. In many ways, it has come to be viewed, along with sickle-cell disease, as a prototypical recessive genetic disorder, one that can teach us a great deal about the molecular basis of disease, population genetics, and delivery of genetic screening services.

Clinical Features

CF is a multisystem disease, affecting a number of different organs and tissues throughout the body. Most of these manifestations have in common the production of abnormally viscous secretions from glands and surface **epithelial cells**. In the lung the mucus secretions from the bronchial epithelial cells are unusually thick. They are difficult to clear properly from the

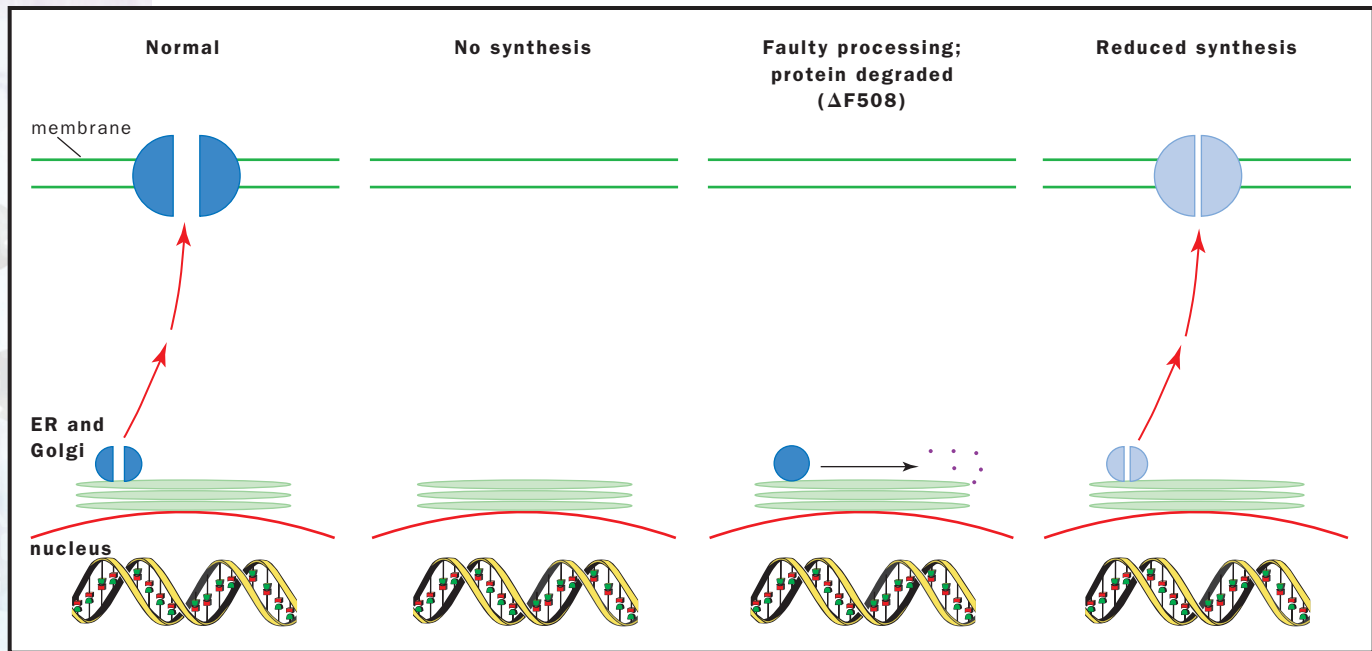
hydrogen bonds weak bonds between the H of one molecule or group and a nitrogen or oxygen of another

template master copy

denaturation destruction of the structure of

epithelial cells one of four tissue types found in the body, characterized by thin sheets and usually serving a protective or secretory function





There are three types of defects in CFTR production that can cause cystic fibrosis. The most common mutation, $\Delta F508$, causes faulty protein processing and degradation of the protein before it reaches the cell membrane. Other defects are also possible.

airway passages and, instead, tend to collect and obstruct the bronchial tree, while providing a perfect culture medium for dangerous bacteria. Over time, repeated bacterial infections damage and destroy the lung tissue, leading to chronic breathing problems and, eventually, to the loss of viable lung function. Indeed, the pulmonary manifestations of the disease are the main cause of death in most CF patients.

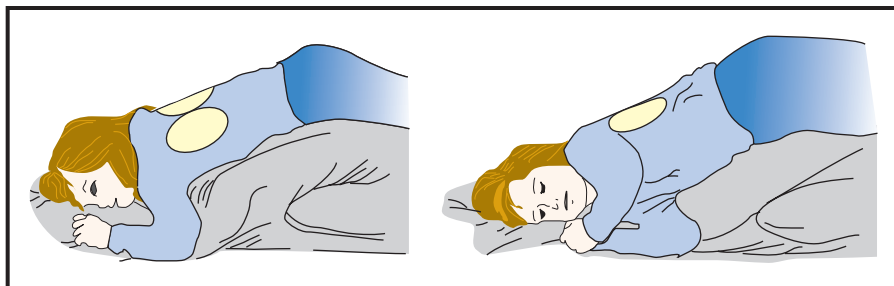
Obstruction in other organs is also seen. In the pancreas it leads to an insufficiency of pancreatic enzymes and malabsorption during digestion; in the nose and sinuses, it produces chronic sinusitis; and in the intestines, in a small minority of newborn infants with CF, it produces an often fatal condition called meconium ileus. Altered secretions also occur in the sweat glands, so that the sweat of CF patients has an abnormally high salt content. In fact, in the early days of medicine, the diagnosis of CF was often made by licking the skin and tasting the sweat! One additional mysterious clinical feature of CF occurs only in men with the disorder: They are infertile due to blockage or congenital absence of the vas deferens, the tube through which sperm pass prior to ejaculation.

Laboratory Diagnosis

Since the clinical symptoms of CF are so varied, diagnosis is aided by laboratory tests. For many years the only definitive test available was sweat chloride analysis, which detects and quantifies the abnormally high salt concentrations in the sweat of CF patients. Since the causative gene was discovered in 1989, as described below, patients can now also be diagnosed by DNA testing, which detects mutations in the gene.

Mode of Inheritance

CF is a classical autosomal recessive disease, in which affected patients are born to parents who are both carriers—that is, they have a mutation in one



Chest physical therapy is the traditional airway clearance technique for cystic fibrosis. It includes postural drainage (positioning to allow movement of secretions) and percussion and vibration on the chest wall to loosen secretions. The circles indicate the sites of percussion. Adapted from <<http://www2medsch.wisc.edu/childrenshosp/CF/cfpages/cpt2.html>>.

of their two CF genes. Carrier couples have a one-in-four risk of producing an affected child with each pregnancy. The carriers themselves are completely asymptomatic and even have normal sweat chloride tests.

Treatment

There is still no cure for this disease, but supportive therapies have improved markedly in the last twenty years. The recurrent lung infections are now much better controlled with powerful, new-generation antibiotics, though, unfortunately, many patients eventually develop infections with drug-resistant bacteria. A variety of approaches are used to break up and remove the thick mucus secretions in the lung, including mucolytic (mucus-breaking) agents, bronchodilators, and chest physical therapy. One type of mucolytic treatment is DNase, a DNA-destroying **enzyme** that breaks up the long, sticky DNA strands left by dying cells. Some patients with end-stage lung disease can be rescued by lung transplantation. Patients with pancreatic obstruction can be managed with pancreatic enzyme supplements, and affected men with vas deferens obstruction have been able to father children by a technique called sperm aspiration, followed by **in vitro** fertilization.

The ultimate hope for the cure of CF lies in gene replacement therapy. A number of clinical trials are under way, most attempting to deliver the normal CF gene to the bronchial epithelium by aerosol spray, using a viral **vector** (usually adenovirus, a common respiratory virus that naturally targets the desired tissue). Thus far the attempts have not been completely successful, as most patients develop an immune response against the virus during the course of therapy. But with the median life expectancy of CF patients now at thirty years just through conventional therapies, the hope is that many CF patients alive today will survive long enough to avail themselves of gene therapy once it is perfected.

The Cystic Fibrosis Gene and CFTR Protein

The identification of the causative gene for CF in 1989 represented one of the great triumphs of molecular genetic research up to that time. With the gene's identification having preceded the official start of the Human Genome Project by one year, the search for the CF gene proceeded without the benefits of the fully mapped genome that we have today. Hence,

enzyme a protein that controls a reaction in a cell

in vitro "in glass"; in lab apparatus, rather than within a living organism

vector carrier

linkage analysis examination of co-inheritance of disease and DNA markers, used to locate disease genes

positional cloning the use of polymorphic genetic markers ever closer to the unknown gene to track its inheritance in CF families

some of the techniques used to identify the CF gene, such as “gene walking” and “gene jumping,” are no longer used extensively.

The CF gene was identified through **linkage analysis** and **positional cloning**. Whereas nowadays the map of the human genome is saturated with these markers, which serve as convenient “signposts” in gene mapping studies, this was not the case when the CF mapping was done, and more laborious, brute-force techniques such as those mentioned above had to be employed. Thus, it was dramatic news indeed when the causative gene was found on chromosome 7.

As might have been expected based on the secretory defects in the disease, the gene, dubbed “cystic fibrosis transmembrane conductance regulator” (*CFTR*), encodes an ion-channel protein in epithelial cell membranes. The gene is quite large—250,000 nucleotides—and the spectrum of mutations in CF patients continues to grow. At the time of this writing, more than 950 different mutations have been reported. Most of these are quite rare and may only be found in individual families. A few are more common, most notably a three-nucleotide deletion of codon 508, called $\Delta F508$, which is found in approximately 70 percent of Caucasian CF carriers. Several others are present in 1 percent to 3 percent of carriers, while the remainder are very rare, except for some that are found at higher frequency in particular ethnic and racial groups (such as W1282X in the Ashkenazi-Jewish population and 3120+1G→A in the African-American population).

The $\Delta F508$ mutation in the *CFTR* gene deletes a phenylalanine amino acid from the final protein. Like other membrane proteins, *CFTR* is made at the endoplasmic reticulum in the interior of the cell, and must be transported to the plasma membrane to function. The absence of this amino acid results in improper folding of the *CFTR* protein within the endoplasmic reticulum, which causes it to be degraded by the cell’s protein-recycling machinery before it reaches the membrane. Some of the less common mutations prevent any protein synthesis by introducing a stop codon into the gene, while others allow the protein to reach the membrane but without functioning properly.

The *CFTR* protein forms a pore to allow chloride ions to pass through the plasma membrane. The full range of functions served by this pore is not known, but the sticky secretions of CF are believed to result when chloride ions in the salty fluid secreted by the epithelial cells cannot be recovered by the membrane protein.

Cystic Fibrosis DNA Testing and Screening

The discovery of the *CFTR* gene raised hopes that the detection of mutations at the DNA level could supplement the traditional sweat test for CF diagnosis and, more importantly, might be used to identify carriers in the general population so that they could be offered genetic counseling. Unfortunately, these goals have been hampered by the large number of possible mutations in the gene, since present-day DNA tests can detect only a small subset of them. As in most recessive diseases, the vast majority of carriers have no family history of the disorder and do not discover that they are

carriers until they happen to have a child with another carrier, giving birth to their first affected child.

CF is an appealing target for population carrier screening simply because of the relatively high carrier frequency in the general population. One in twenty-nine Caucasians, one in forty-six Hispanics, one in sixty-five African Americans, and one in ninety Asian Americans are carriers. It is not known why the mutation frequency is so high, especially in European populations. Some have proposed, using the analogy of the sickle-cell gene conferring relative resistance to malaria, that the mutations must have a protective effect against some disease appearing in European history, such as cholera or tuberculosis.

But all of this is just speculation. DNA screening of the entire adult population could potentially identify those couples at risk, who could then be offered prenatal diagnosis, affording couples the opportunity to consider their options. After several pilot studies and much debate at the national level, it has now been recommended that screening for the twenty-five most frequent *CFTR* mutations be offered to all couples expecting a child or planning a pregnancy. So most of the students reading this book will eventually be offered this DNA test! SEE ALSO CELL, EUKARYOTIC; GENE DISCOVERY; GENE THERAPY; GENETIC COUNSELING; HETEROZYGOTE ADVANTAGE; HUMAN DISEASE GENES, IDENTIFICATION OF; INHERITANCE PATTERNS; POPULATION SCREENING; PROTEINS.

Wayne W. Grody

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Delbrück, Max

Physicist, Molecular Biologist
1906–1981

Max Delbrück made major contributions to the understanding of replication and viral function. Raised in Berlin in a distinguished family of German intellectuals, Delbrück trained as a physicist with Niels Bohr and other leaders in the field of quantum mechanics. In the early 1930s his interests turned toward biology and the nature of the gene. This was only thirty years after the rediscovery of Mendel's work and twenty years before Watson and Francis Crick discovered the structure of DNA. With two colleagues, he published a theoretical paper on quantum mechanical restrictions on gene structure. These ideas were popularized by the physicist Erwin Schrödinger in the book *What is Life?*, which inspired many young midcentury scientists to join the quest to understand the gene.

D



Max Delbrück.

bacteriophages viruses
that infect bacteria

Delbrück moved to the United States in 1937 to pursue genetics and escape the increasingly repressive atmosphere of Nazi Germany. He first went to Columbia University in New York, where he joined Thomas Hunt Morgan's group to study *Drosophila*. Soon, however, he became interested in **bacteriophages**. It was in the understanding of this model system that Delbrück made his greatest contribution.

Bacteriophages are among the simplest genetic systems, and thus provided Delbrück with an elegant tool for exploring fundamental processes of reproduction and mutation. Delbrück collaborated with Salvador Luria and Alfred Chase to work out the fundamental mechanisms of viral replication and to explore the genetics of mutation in this system. This loosely allied trio, and the ever-widening circle of scientists with whom they collaborated, became known as "the phage group." This group conducted training courses at Cold Spring Harbor Laboratory in New York, where they introduced many other biologists to this model system, while inculcating in them their own rigorous and quantitative approach. Watson was one of Delbrück's students in the phage course. The phage group began and shaped the field of molecular genetics, and Delbrück is usually considered the father of this discipline. Delbrück, Luria, and Hershey were awarded the Nobel Prize for physiology or medicine in 1969 for their discoveries in phage genetics.

Later in his life, Delbrück turned his attention to the cellular physiology underlying perception, but the model system he chose for this research, a light-sensitive fungus, had too little in common with animals to make the research strongly relevant to animal perception. He died in 1981. SEE ALSO CRICK, FRANCIS; MORGAN, THOMAS HUNT; WATSON, JAMES; VIRUS.

Richard Robinson

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Development, Genetic Control of

Development is the process through which a multicellular organism arises from a single cell. During development, cells become specialized, or differentiated, taking on different functions and forms. The organism develops a characteristic three-dimensional shape, the parts of which (such as limbs and organs) continue to maintain the same relationship to each other even as the organism grows. How the genes in a single fertilized egg dictate the creation of a complex multicellular creature is the central question in the genetic control of development.

While we are often most curious about human developmental processes, very little is known about the genetics of human development specifically, because experimentation on human embryos is forbidden by law and ethics. Instead, the details of genetic control are best understood in several model organisms, including the roundworm (*Caenorhabditis elegans*), the fruit fly

(*Drosophila melanogaster*), the zebrafish, and the mouse. Each organism differs in the details, and in some cases the overall logic, of genetic control. The understanding of developmental control is not complete for any of these organisms, but scientists have come to understand several mechanisms that contribute to, but do not entirely explain, development.

The Importance of Transcription Factors

With few exceptions, every cell in a multicellular organism contains the same set of genes as every other cell. Despite this genetic equivalence, cells differ greatly in form, function, longevity, and many other characteristics. These differences are due to the differential expression of genes within each cell type. Thus, a nerve cell will express a certain subset of the entire **genome**, while a gut cell will express a different subset. (To express a gene means to use it to create its encoded product, usually a protein.) Cells become different from one another, therefore, by expressing different sets of genes. Thus, the problem of development can be addressed by understanding how initially identical cells come to express different sets of genes.

The beginning of the answer to this question lies in understanding gene transcription and transcription factors. Transcription is the process in which an enzyme called RNA polymerase binds to a gene to make an RNA copy; this is the first step in expressing the gene. Transcription factors are proteins that bind to regulatory regions of the gene, thereby influencing how easily RNA polymerase attaches to it. Different genes require different sets of transcription factors, and when these factors are in low supply, expression of that gene is slowed or stopped.

Since transcription factors are proteins, they are encoded by their own genes, which are regulated by yet other transcription factors. As we will see, many of the “master” genes that control development encode transcription factors that are expressed early in development. The sequential activation of these genes, in a domino-like fashion, is one way that the overall developmental program is carried out.

The European Way and the American Way

A central question in development is whether a particular cell is predestined to become a specific cell type from the moment of its creation, or whether its fate is less determinate, depending on a variety of cues it receives from its local environment as development proceeds. The developmental geneticist Sydney Brenner dubbed these two alternatives the European way (what matters is who your ancestors are) and the American way (what matters is what your environment is and who your neighbors are). While no multicellular organism displays either alternative exclusively, the roundworm *C. elegans* operates primarily according to the European plan, with each of its exactly 959 cells largely following a set developmental path, and with few decisions made through interaction with neighboring cells. *Drosophila*, and mammals such as mice and humans, largely develop according to the American plan, with most cells only gradually taking on a final identity, through repeated communication and competition with neighboring cells.

genome the total genetic material in a cell or organism



The difference can be seen in transplant experiments, in which cells of the early embryo are moved from one region to another. In the roundworm, the transplanted cell generally follows its original developmental plan, regardless of the environment in which it finds itself. In fruit flies and mammals, the transplanted cell generally takes on the identity of the region into which it is transplanted, switching from one developmental pathway to another, such as from bone cell to gut cell. This change is not absolute and, most importantly, it is time-dependent. Cells transplanted later in development tend to remain committed to the pathway they were on, despite their new surroundings, leading to the aberrant development of bone cells in the gut, for instance.

Morphogens and Gradients

One problem has intrigued embryologists for many decades: In the absence of strictly defined developmental fates, how does a cell “know” where it is in an embryo, in order to know what to become? An early suggestion, which has been borne out by experiments, is based on the concept of a concentration gradient—a variation in concentration of a substance across a region of space.

A concentration gradient is formed whenever a substance is created in one place and moves outward by diffusion. When this occurs, there will be a high concentration of the substance near its point of origin, and increasingly lower concentrations further away. This provides positional information to a cell anywhere along the gradient. Cells pick up this chemical signal, and its strength (concentration) determines the cell’s response. Typically, the signal is a transcription factor, and the response is a change in gene transcription.

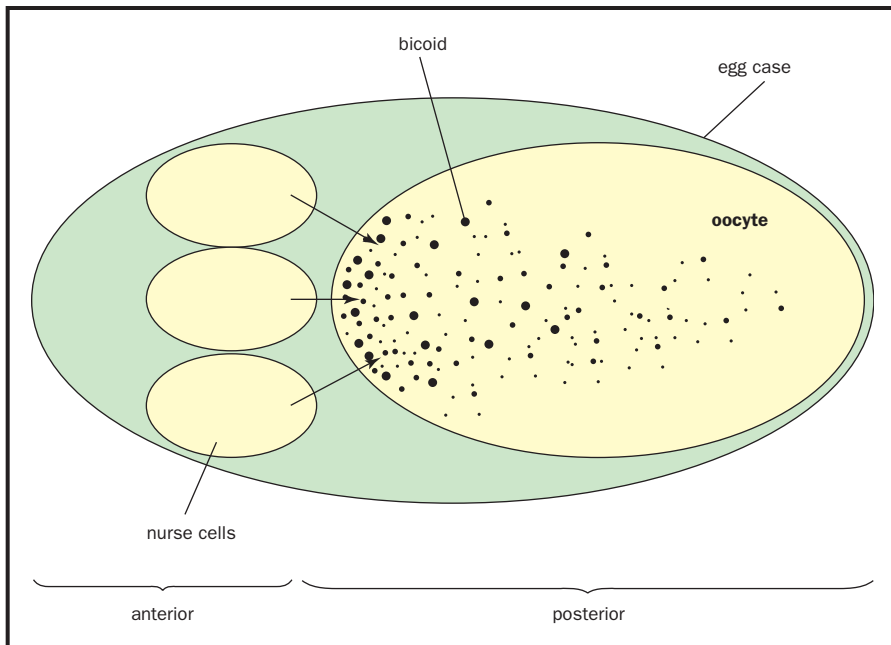
Because such a signalling substance helps to give form to the embryo, it is termed a morphogen (“morph-” meaning “form,” “-gen” meaning “to give rise to”). Morphogen gradients are a key means by which originally identical cells are exposed to different environments and thus sent along different developmental paths. In humans and other mammals, one morphogen that acts early in development is retinoic acid, a relative of vitamin A.

Retinoic acid is used as an acne treatment for humans, and must be avoided during pregnancy.

Gradients Determine the Axes of the Fruit Fly Embryo

We can see how such a morphogen acts by considering the development of the anterior-posterior axis in the fruit fly embryo. In the fly egg case, the oocyte, or fertilized egg, is accompanied by “nurse cells” at what will become the head end of the fly. This is called the anterior end; the tail end is posterior. Nurse cells create messenger RNA for a protein called bicoid, which they transport to the oocyte. Because these **mRNAs** originate in the anterior end, their concentration is highest there, and is lower towards the posterior end. Once the oocyte begins to divide, the mRNA is translated, and the bicoid protein is synthesized. Anterior cells have more of it than posterior cells, and the difference in concentration sets each cell group down its own developmental pathway, with anterior cells developing head structures, and posterior cells tail structures. Note that, in keeping with the “American plan,” the fate of each cell is determined not by its ancestry, but by the environment it is in.

mRNA messenger RNA



A concentration gradient of bicoid mRNA (and later bicoid protein) helps establish the anterior-posterior axis within the fruit-fly oocyte.

The effect of bicoid can be seen in transgenic flies, which have too many or too few copies of the gene. With extra bicoid, a higher-than-normal concentration exists further back in the oocyte, and anterior structures develop further back on the fly. With no bicoid, the anterior structures don't develop at all.

As we might expect, the bicoid protein is a **transcription factor**, which helps regulate expression of other genes. Other gradients of other transcription factors also exist at this stage, and together, these overlapping gradients establish the dorsal-ventral (back-belly) axis and map out the body segments that characterize all insects. While the details are complex, the fundamental idea is that of combinatorial control: At each position, it is the combination of transcription factors and their concentrations that determines which genes will be expressed, and therefore what the identity of the cells will be.

transcription factor
protein that increases the rate of transcription of a gene

As segmentation becomes more firmly established and segments begin to take on their unique identities, gradients become less important. Instead, local gene control and cell to cell interactions create the increasingly fine level of spatial patterning.

Homeotic Genes and Segment Identity

Once segmentation is established, another important and remarkable set of genes turns on: the homeotic selector genes. These genes control development within each segment. For instance, the thorax region of the fly contains three segments, each with one pair of legs (the reason insects are six-legged). The homeotic selector gene *antennapedia* is normally expressed only in the thoracic segments, leading to the creation of a pair of legs.

Note that *antennapedia* does not itself “code for” legs. Instead, its protein product is a transcription factor. By regulating expression of many other